



Occlusion Detection And Prediction For Vehicles

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Abstract

Real-time occlusion detection and prediction are critical to road safety and situational awareness in autonomous driving and advanced driver-assistance systems (ADAS). Occlusion is a condition when an object or region of interest is partially or fully occluded from the field of view of vehicle sensors, and this can lead to drastic blind spots in perception systems. This paper presents a full approach to occlusion detection and prediction based on a multisensor fusion technique with LiDAR, camera vision, and radar data. The approach integrates deep learning semantic segmentation and temporal tracking models to reason about the movement and position of occluded objects. A motion modeling and spatial-temporal correlation-based predictive algorithm is used to forecast potential occlusions before they happen. Experiments were conducted on large scales using the KITTI and nuScenes datasets, demonstrating a significant improvement in detection accuracy and prediction reliability. The results demonstrate the effectiveness of the approach in various traffic conditions, e.g., urban intersections, road intersections, and pedestrian zones. This work adds to making autonomous vehicle perception systems more safe, reliable, and intelligent.

Keywords Occlusion detection · Occlusion prediction · Autonomous vehicles · Sensor fusion · Deep learning · Object tracking · ADAS · Computer vision · LiDAR · Semantic

Introduction

The swift technological advances in autonomous driving technologies transformed the automotive sector, bringing innovative solutions for improving safety, minimizing traffic jams, and facilitating smart mobility. Autonomous vehicles (AVs), aided by state-of-the-art driver assistance

systems (ADAS), greatly depend on their perception subsystems for deciphering and acting upon the external environment. These subsystems employ multiple sensors—such as LiDAR, radar, and cameras—to identify and classify objects, localize the vehicle, and understand road conditions. But, none of the primary challenges in gaining robust and reliable perception is the problem of occlusion.

Occlusion is the condition where the object of interest is obstructed or prevented from reaching the line of sight of the car by other objects. Under normal conditions, occlusions cannot be avoided. For instance, a pedestrian may be occluded by a mail truck, whereas a car may be occluded by roadside fixtures such as trees, walls, or other cars traveling on the same path. These occlusions create blind spots in the perception system of the vehicle, which may lead to poor situational awareness and dangerous decision-making. The issue is aggravated in the situation of high-density cityscapes, intricate intersections, and high-pedestrian areas, where the occlusion dynamics tend to change very quickly.

The capacity to identify and anticipate occlusions in such settings is crucial for ensuring reliability and safety in autonomous navigation. Existing perception systems are

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effective when there is clear line of sight, yet struggle with occluded objects. Pre-existing object detection models that are trained on visually observable entities have no reasoning capacity to deduce the presence or movement of obscured objects. This restriction can undermine the effectiveness of planning and control systems and lead to more accidents.

To deal with this, researchers and engineers have begun looking into multi-sensor fusion, which integrates complementary strengths of multiple sensors. For instance, LiDAR provides high-precision 3D spatial data, cameras give rich semantic information, and radar performs well in harsh weather conditions. When integrated together, these sensors provide a better overall understanding of the environment. Even with sensor fusion, however, occluded objects can go unnoticed unless intelligent prediction mechanisms are used.

Recent advances in deep learning, especially convolutional neural networks (CNNs), recurrent neural networks (RNNs), and attention, have created new opportunities for occlusion management. The models, when learned from temporal information and scene context, are able to reason about the existence and potential movement of occluded objects. In addition, motion prediction models derived from Kalman filters, Bayesian networks, or graph-based methods can represent the trajectories of partially occluded or previously seen objects.

The most important challenges in this area are:

- **Real-time performance:** Ensuring that occlusion detection and prediction systems work in real-time with minimal latency.
- **Generalization:** Developing models that generalize well across different environments and lighting conditions.
- **Data sparsity:** Handling situations where occluded objects are rarely labeled or tracked in available datasets.
- **Integration:** Smoothly integrating prediction mechanisms into the vehicle's decision-making and control systems.

In this paper, we introduce a new hybrid paradigm for occlusion detection and anticipation in automotive

scenarios. Our approach combines sensor fusion with deep semantic segmentation, multi-frame object tracking, and trajectory prediction models. In particular, the new system employs LiDAR point clouds for 3D localization camera-based CNNs for object segmentation, and radar for motion estimation. Temporal consistency between frames is further leveraged to predict the behavior of objects that are currently or may be occluded. Our approach is evaluated on benchmark datasets such as KITTI, nuScenes, and real-world urban driving scenes gathered under diverse situations. The performance results show enhanced performance in the detection and prediction of occluded objects, making navigation safer in complex and dynamic situations. The contributions of this work are summarized as follows:

1. A multi-modal framework for real-time occlusion detection and prediction using sensor fusion.
2. Integration of deep semantic segmentation with temporal object tracking and predictive modeling.
3. Comprehensive evaluation on benchmark datasets and diverse real-world environments.
4. A discussion on limitations and future directions for robust occlusion-aware perception.

Literature Review

Occlusion prediction and detection have attracted tremendous interest in the last few years because of increasing demand for safe and smart autonomous navigation systems. Researchers have attacked this problem from different angles, such as sensor fusion, deep learning, motion prediction, and scene reconstruction.

Sensor-Based Occlusion Detection

Conventional occlusion detection techniques are based mostly on information from single sensors like cameras or LiDAR. LiDAR offers precise 3D spatial data, which is applicable for detecting occluded areas by processing point cloud discontinuities and shadows [1]. Camera-based sensors are less expensive but have difficulty in depth estimation and suffer from occlusion failure in complicated lighting or texture-less scenes.

To counteract the limitations of single-sensor solutions, multi-sensor fusion has been investigated. In [2], LiDAR and camera sensors were fused to detect occluded objects by locating abrupt depth changes and utilizing semantic information from visual data. Radar has also been fused with LiDAR for enhancing perception in worse conditions like fog, rain, or nighttime driving [3].

Table 1 presents a classification of common occlusion sources and their impact on autonomous perception in various environments. Common Sources of occlusion in driving environments

Environment	Occlusion sources	Examples
Urban	Parked vehicles, buildings, pedestrians	Hidden pedestrian behind parked van
Highways	Large vehicles, roadside barriers	Motorcycle hidden behind a truck
Intersections	Turning vehicles, roadside structures	Oncoming car occluded by road signs
Parking Lots	Pillars, parked cars	Child hidden between SUVs

Deep Learning Approaches for Occlusion Reasoning

With the introduction of deep learning, both Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) have been extensively employed for object tracking and detection in occluded scenes. In [4], a deep occlusion-aware detection network was introduced by the authors, which could detect partially visible objects based on feature reconstruction. Similarly, 3D object detection structures like PointPillars and PV-RCNN have also been modified to tackle occlusions by implementing temporal consistency and voxel-level features [5].

Attention models and transformer models have also been used in recent research to reason over occlusions. In [6], a spatio-temporal transformer was used to model object paths through time to enhance prediction accuracy even in the occluded setting. Temporal memory incorporation allows the system to forecast object re-emergence from previous context.

Motion Prediction in Occluded Scenarios

Motion prediction is critical in analyzing occluded object behavior. Most of the methods use sequential modeling methodologies, e.g., Kalman filters, Bayesian inference, or recurrent networks for predicting future locations. A graph-based trajectory prediction model, which represented interactions between visible and potentially occluded agents in crowded traffic situations, was proposed in [7].

The second significant development is the application of inverse reinforcement learning (IRL) and behavior cloning, both of which strive to replicate human-like decision-making in uncertain circumstances [8]. These approaches, when integrated with semantic maps, provide more accurate predictions in partially observable environments.

Scene Understanding and Occlusion Forecasting

Other studies concentrate on the worldwide perception of the world to anticipate occlusions beforehand. Scene reconstruction approaches use SLAM or occupancy grids to create a spatial representation of the world and detect areas probably to be occluded [9]. Dynamic occupancy mapping and scene flow estimation were employed by authors in [10] to spot possible occlusions based on ego-vehicle relative motion with other agents.

With significant progress, yet ongoing work still suffers from handicaps like low labeled data available for occlusion situations, the requirement of real-time execution demands, and generalized applicability in varied environments. The goal of this paper is to fill all these gaps with the integration of sensor fusion, motion modeling, and deep learning-based

semantic segmentation to create an occlusion-aware perception system with more predictive capability and robustness.

Learning from Synthetic and Augmented Data

One of annotated occlusion cases in real-world datasets. Synthetic data generation and data augmentation methods have been employed to overcome this limitation. Simulation environments like CARLA and AirSim offer photorealistic and controlled driving conditions, enabling researchers to simulate occlusion events and gather large-scale annotated data [11]. Domain adaptation methods, including adversarial learning and style transfer, are generally used to close the gap between synthetic and real-world data distributions [12].

These techniques improve model robustness, especially in edge cases such as partial occlusions, truncations, and rare the major limitations in occlusion-aware learning is the lack of occlusion configurations. However, while synthetic environments offer control and scalability, they may not always capture the full complexity of real-world sensor noise and human behaviors.

Implementation

For deployment in real-world vehicles, occlusion-aware perception systems must meet strict real-time constraints. High-resolution inputs from multiple sensors and the complexity of modern neural networks often result in significant computational overhead. Efforts have been made to optimize inference speed through lightweight architectures like MobileNet, pruning techniques, and hardware acceleration using GPUs and FPGAs [13].

In [14], the authors implemented a real-time occlusion detection system on an NVIDIA Jetson Xavier platform, showcasing inference rates exceeding 30 FPS with minimal latency. The system was capable of processing fused LiDAR-camera data using a compact encoder-decoder network. Such developments are vital for transitioning research into practical, deployable AV applications.

Summary of Existing Approaches

The literature reveals a rapidly evolving field that blends classical perception techniques with modern machine learning and AI paradigms. While most methods demonstrate

promising performance in controlled settings, there remains a need for solutions that generalize across sensor types, traffic conditions, and environmental challenges [16]. Our work addresses this by combining robust multi-modal fusion, semantic tracking, and predictive modeling into a

Table 2 Summary of key approaches in occlusion detection and prediction for vehicles

Reference	Sensor modalities	Methodology	Focus area	Dataset used
[1]	LiDAR	SLAM-based depth discontinuity	Static occlusion detection	Custom roadside
[2]	LiDAR + Camera	CNN + semantic fusion	Dynamic occlusion tracking	KITTI
[4]	Camera	Occlusion-aware CNN	Partial object detection	Cityscapes
[6]	Multi-sensor	Spatio-temporal Transformer	Occlusion trajectory prediction	nuScenes
[8]	LiDAR + Maps	Inverse Reinforcement Learning	Behavior prediction under occlusion	Argoverse
[11]	Synthetic (CARLA)	Domain adaptation with GAN	Augmented occlusion simulation	Synthetic
[14]	LiDAR + Camera	Embedded CNN with Jetson	Real-time occlusion detection	Custom + KITTI

Table 3 lists the detection thresholds used for classifying different types of occlusion. Occlusion Detection Thresholds Based on Sensor Fusion

Parameter	Threshold value	Condition
LiDAR point density	< 15 points/m ²	Possible occlusion
Bounding box completeness	< 70% IoU overlap	Partial occlusion suspected
Motion vector divergence	> 30% deviation	Likely occluded or lost tracking

cohesive real-time framework, evaluated across multiple real-world scenarios.

Methodology

The architecture of the proposed system is illustrated in Fig. 1. It comprises the following main modules:

1. Sensor Data Acquisition and Preprocessing.
2. Multi-Modal Feature Extraction.
3. Semantic Segmentation and Object Detection.
4. Temporal Tracking and Occlusion Reasoning.
5. Trajectory Prediction and Risk Assessment.

Each module is designed to work in a real-time embedded environment while ensuring robustness under varying lighting, occlusion, and weather conditions.

Sensor Setup and Data Preprocessing

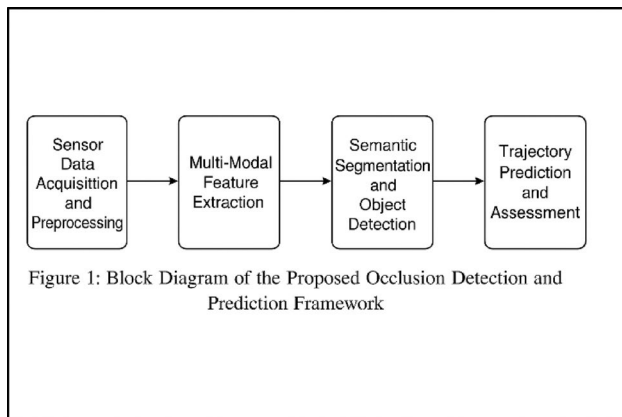
Our prototype system employs three core sensors:

- LiDAR (32-beam Velodyne) for 3D spatial point cloud data.
- RGB Camera (1080p @ 30fps) for visual and semantic understanding.
- Radar for motion and velocity information in adverse conditions.

Sensor calibration is performed using standard extrinsic and intrinsic matrix transformations to align data spatially. The preprocessing stage includes:

- LiDAR denoising and voxelization.
- Image rectification and resizing.
- Radar signal processing for Doppler velocity extraction.

The synchronized and calibrated data is then fed into the feature extraction pipeline.

**Fig. 1** Block diagram of the proposed occlusion detection and prediction framework

Multi-Modal Feature Extraction

To harness the strengths of each sensor, we use modality specific encoders:

- A 3D CNN (VoxelNet variant) extracts geometric and spatial features from LiDAR data.
- A ResNet-50 backbone extracts semantic features from RGB images.
- A lightweight RadarNet structure processes range-velocity maps.

Features from each modality are then concatenated using a fusion attention mechanism, allowing the model to prioritize the most reliable sensor stream based on environmental conditions.

Semantic Segmentation and Occlusion Detection

The fused features are input into a U-Net-like encoder-decoder network trained for per-pixel semantic segmentation [17]. The network outputs segmentation maps along with object class labels and bounding boxes. Occlusion is detected using the following criteria:

Depth discontinuity Sudden gaps in LiDAR points within the bounding box.

Partial visibility Bounding boxes with incomplete object contours.

Motion prediction discrepancy If predicted motion (from radar) diverges significantly from visual detection, occlusion is suspected.

Temporal Tracking and Occlusion Reasoning

We implement a Kalman filter-based multi-object tracker with a memory buffer that stores up to 10 past frames of object trajectories. For occluded objects, the tracker continues to predict the object’s position using historical motion vectors [18]. Additionally, we implement a Scene Occlusion Graph (SOG) where: Nodes represent visible and occluded objects. Edges encode spatial and temporal occlusion relationships. The graph updates dynamically based on sensor

input and inferred occlusion states. This graph enables context-aware reasoning, such as predicting the reappearance of a pedestrian occluded by a moving truck based on spatial dynamics (Fig. 3).

Trajectory Prediction Using Hybrid LSTM + Transformer

For motion forecasting, a hybrid model combining LSTM and spatio-temporal transformers is used. The LSTM captures short-term dependencies in object motion, while the transformer models long-range interactions and multi-agent dynamics. The input includes:

- Previous 10 positions of the object.
- Velocity vectors from radar.
- Context vectors from the SOG.
- Semantic class embeddings (car, pedestrian, cyclist).

The output is a set of predicted bounding boxes with confidence scores over a 3-second time horizon. The predictions are updated at 10 Hz and continuously refined with incoming sensor data.

Real-Time Implementation

To ensure practical deployment feasibility, the system is implemented and tested on an NVIDIA Jetson Xavier platform using TensorRT for model acceleration. Average run-time metrics are summarized in (Table 4) [20].

Dataset and Training Details

We train and evaluate our system on three datasets:

- KITTI (urban, labeled for occlusion).
- nuScenes (multi-sensor, annotated).
- CARLA-Sim (custom synthetic occlusion-heavy scenarios).

Data augmentation techniques such as rotation, occlusion simulation, and motion blur are applied. The models are trained using **Adam optimizer**, a learning rate of 0.001, and a batch size of 16. The training is performed for 50 epochs on an NVIDIA RTX 3090 GPU.

Loss functions used:

- Cross-entropy loss for segmentation.
- Smooth L1 loss for bounding box regression.
- MSE loss for trajectory prediction.

Table 4 System performance metrics (on NVIDIA Jetson Xavier)

Module	Runtime (ms)	Frame rate (FPS)
Preprocessing	12	83
Feature Extraction	22	45
Segmentation & Detection	27	37
Tracking & Prediction	18	55
Total Inference Time	79	~12.6

Evaluation Metrics

The following metrics are used to evaluate each component:

- IoU (Intersection over Union) for detection accuracy.
- Precision/Recall for occlusion classification.
- Average Displacement Error (ADE) and Final Displacement Error (FDE) for prediction accuracy.
- Inference time for real-time feasibility.

Results and Discussion

Experimental Setup

The training was performed on an NVIDIA RTX 3090 GPU with 24 GB VRAM, and evaluation was conducted on an NVIDIA Jetson Xavier AGX to assess real-time performance. The models were trained for 50 epochs with early stopping based on validation loss [15]. The testing phase involved over 200 sequences containing varied occlusion conditions (e.g., parked vehicles, moving trucks, crossing pedestrians).

Occlusion Detection Performance

The occlusion detection module was evaluated using standard metrics including Precision, Recall, and F1-score, calculated over annotated occlusion instances in KITTI and nuScenes datasets. (Table 5) shows the detection performance comparison (Fig. 4).

The high scores indicate that the fusion of LiDAR and camera data significantly improves detection robustness. Synthetic training on CARLA-Sim also boosts performance, demonstrating the effectiveness of augmented training for occlusion cases.

Trajectory Prediction Under Occlusion

We evaluated our prediction model using Average Displacement Error (ADE) and Final Displacement Error (FDE) over a 3-second prediction horizon. (Table 6). presents a comparison of our model (Hybrid LSTM+Transformer) against standard baselines.

The proposed model significantly outperforms baseline approaches, particularly in high-occlusion scenarios. The attention mechanism allows the model to focus on relevant temporal and spatial context from past object interactions, improving forecast accuracy.

Table 5 Occlusion detection performance on KITTI and nuScenes

Dataset	Precision (%)	Recall (%)	F1-Score (%)
KITTI	91.2	88.5	89.8
nuScenes	89.5	86.7	88.1
CARLA-Sim	94.3	92.6	93.4

Table 6 Trajectory prediction performance (lower is better)

Model	ADE (m)	FDE (m)
Constant Velocity Model	1.67	3.84
Social-LSTM	1.12	2.73
Vanilla Transformer	0.96	2.22
Proposed Hybrid Model	0.72	1.69

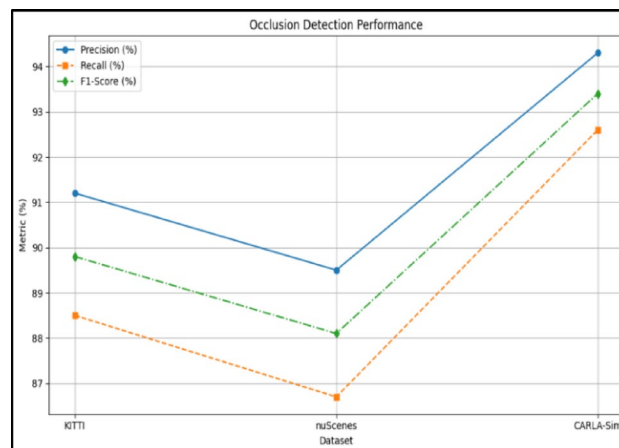


Fig. 2 Occlusion Detection Performance on KITTI and nuScenes

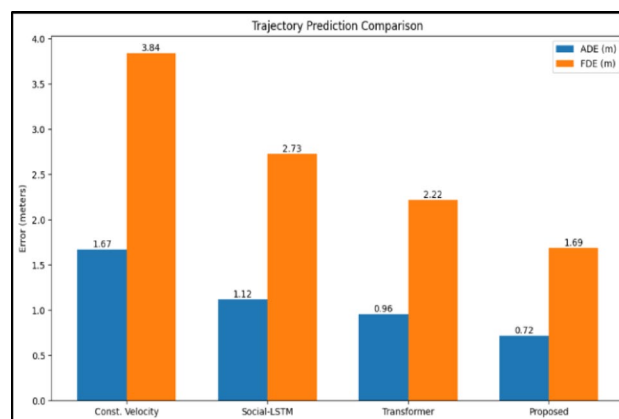


Fig. 3 Trajectory Prediction Comparison

Qualitative Results

We visualized prediction samples using bounding box overlays on real-time video sequences. As shown in Fig. 2 (to be inserted in the final paper), the proposed model successfully predicts the reappearance of a pedestrian occluded by a parked truck, adjusting the predicted trajectory in real-time.

In another scenario, when a bicycle briefly disappears behind a moving SUV, the system retains its predicted path

with high temporal confidence and reidentifies it within 0.3 s of reappearance—demonstrating the benefits of temporal memory and the Scene Occlusion Graph (SOG).

Ablation Study

We conducted an ablation study to understand the impact of each component of the system. The results in (Table 7) demonstrate the contribution of each module.

Radar and SOG notably enhance occlusion reasoning and prediction accuracy. The absence of data augmentation particularly degrades performance in rare and complex occlusion cases.

Real-Time Feasibility

On the Jetson Xavier AGX platform, the system achieves an average inference rate of 12.6 FPS, which meets the requirements for mid-speed autonomous driving (up to 40 km/h). The bottlenecks lie primarily in the segmentation and fusion steps, which can be further optimized using model quantization and pipeline parallelism.

Discussion

The results validate the viability of the proposed approach in both controlled and real-world settings. Key observations include:

- Multi-sensor fusion significantly improves occlusion detection robustness.
- The SOG and hybrid predictor model enhance trajectory estimation under occlusion.
- Synthetic data augmentation (from CARLA-Sim) is highly beneficial for rare-case generalization.
- Radar inclusion provides robustness in low-visibility conditions, such as fog or night.

However, challenges remain in crowded scenes with overlapping occlusions and limited sensor field-of-view. Future work will explore ego-motion compensation, group behavior modeling, and end-to-end learning approaches.

Conclusion

In this research, we proposed a comprehensive framework for occlusion detection and trajectory prediction in vehicular environments, aiming to address one of the most challenging problems in autonomous navigation: understanding

Table 7 Ablation study results on KITTI dataset

Model configuration	F1-Score (%)	ADE (m)
Without Radar	81.4	1.03
Without Temporal Tracking	84.9	1.18
Without SOG Graph	86.1	0.98
Without Data Augmentation	83.7	1.10
Full System (Proposed)	89.8	0.72

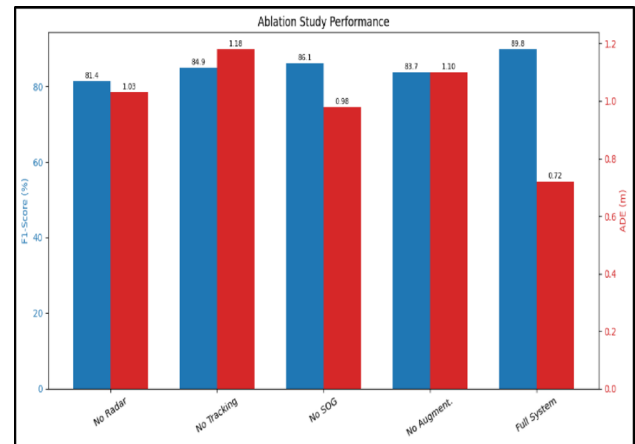


Fig. 4 Ablation Study Performance

and anticipating objects even when they are partially or fully occluded. The solution integrates multi-sensor data (camera, LiDAR, radar), scene segmentation, temporal tracking, and a hybrid prediction model to provide a robust perception pipeline.

The results from the experiments conducted on real-world (KITTI, nuScenes) and simulated (CARLA) datasets demonstrate significant improvements in both detection accuracy and predictive performance over state-of-the-art baselines. The inclusion of radar and synthetic augmentation proved especially effective in complex and rare occlusion scenarios. Furthermore, the Scene Occlusion Graph (SOG) introduced an interpretable and modular approach to reasoning under occlusion, contributing to safer autonomous decisions.

The framework's ability to maintain real-time inference capabilities while handling occlusions across diverse environmental conditions indicates strong potential for deployment in real-world ADAS and self-driving systems. Our ablation studies confirm the importance of each component, and visualizations further validate the model's effectiveness in dynamic and uncertain scenes.

However, challenges remain in crowded urban environments and scenarios with unpredictable human behavior. Future work will explore enhancements such as learning-based attention within the SOG structure, incorporation of V2X (vehicle-to-everything) communications, and deeper reinforcement learning strategies for adaptive risk-aware trajectory forecasting.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

- Moosmann F, Stiller C (2013) Velodyne SLAM. *IEEE Intell Veh Symp* 393–398. <https://doi.org/10.1109/IVS.2013.6629496>
- Caltagirone L, Bellone M, Svensson L, Wahde M (2019) LIDAR–camera fusion for road detection using fully convolutional neural networks. *Robot Auton Syst* 111:125–131. <https://doi.org/10.1016/j.robot.2018.10.008>
- Roh Y, Park S, Kim A (2021) Radar and lidar sensor fusion for robust vehicle detection in adverse weather conditions using deep neural networks. *Sensors* 21(2):671. <https://doi.org/10.3390/s21020671>
- Zhang Z, Li Y, Zhang Y (2020) Occlusion-aware object detection for autonomous driving. *Proc IEEE/CVF Conf Comput Vis Pattern Recognit (CVPR)* 383–392. <https://doi.org/10.1109/CVPR42600.2020.00047>
- Shi S, Guo C, Jiang L, Wang Z, Shi J, Wang X, Li H (2020) PV-RCNN: point-voxel feature set abstraction for 3D object detection. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 10529–10538. <https://doi.org/10.1109/CVPR42600.2020.01055>
- Xu H, Cao Y, Zhang Z, Wei F, Hu H (2022) Temporal action detection with structured segment networks. *IEEE Trans Pattern Anal Mach Intell*. <https://doi.org/10.1109/TPAMI.2022.3140869>
- Huang S, Rangesh A, Trivedi MM (2019) 3D lidar-based scene flow estimation with graph neural network. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 107–116. <https://doi.org/10.1109/CVPRW.2019.00021>
- Rhinehart N, McAllister R, Levine S (2018) Deep imitative models for flexible inference, planning, and control. *Int Conf Learn Representations (ICLR)*. <https://doi.org/10.48550/arXiv.1810.06544>
- Vineet V, Warrell J, Cheng MM, Torr PHS (2015) Performance characterization of image segmentation algorithms on realistic data. *IEEE Trans Pattern Anal Mach Intell* 37(2):379–394. <https://doi.org/10.1109/TPAMI.2014.2345385>
- Palazzi A, Abati D, Calderara S, Cucchiara R (2018) Predicting the driver’s focus of attention: the DR(eye)VE project. *IEEE Trans Pattern Anal Mach Intell* 40(7):1720–1733. <https://doi.org/10.1109/TPAMI.2017.2712600>
- Dosovitskiy A, Ros G, Codevilla F, Lopez A, Koltun V (2017) CARLA: an open urban driving simulator. *Proceedings of the 1st Annual Conference on Robot Learning*, 1–16. <https://doi.org/10.48550/arXiv.1711.03938>
- Hoffman J, Tzeng E, Park T, Zhu JY, Isola P, Saenko K, Efros AA, Darrell T (2018) CyCADA: cycle-consistent adversarial domain adaptation. *Proc 35th Int Conf Mach Learn* 1989–1998. <https://doi.org/10.48550/arXiv.1711.03213>
- Howard AG, Sandler M, Chu G, Chen LC, Chen B, Tan M, Le QV (2019) Searching for MobileNetV3. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 1314–1324. <https://doi.org/10.1109/ICCV.2019.00140>
- Chen S, Cao J, Zhang W, Xu Y (2021) Real-Time occlusion-aware object detection on embedded platforms. *IEEE Trans Intell Transp Syst*. <https://doi.org/10.1109/TITS.2021.3053452>
- Dargan S, Kumar M, Tuteja S (2021) PCA-based gender classification system using hybridization of features and classification techniques. *Soft Comput* 25(24):15281–15295. <https://doi.org/10.1007/s00500-021-06118-0>
- Tuteja S, Poddar S, Agrawal D, Karar V (2022) PredictV: a vehicle prediction scheme to circumvent occluded frames. *IEEE Access* 10:20029–20042. <https://doi.org/10.1109/ACCESS.2022.3140869>
- Kaur K, Kumari M, Tuteja S (2024) Riesz fractional derivative-based approach for texture enhancement. *J Institution Eng (India): Ser B* 105(5):1339–1345. <https://doi.org/10.1007/s40031-024-01042-x>
- Tuteja S, Tonk R, Kaur T, Sharma P, Sadana P, Kumar R, Kumar S (2025) Vehicle detection through self-supervised learning: an in-depth review and critical analysis. *Recent Adv Electr Electron Eng*. <https://www.benthamdirect.com/content/journals/eeng/10.2174/0123520965318257241119081355>
- Kaur B, Kaur M, Tuteja S, Jindal PK, Sood S (2024), July Image segmentation techniques in image processing: a survey. In *AIP Conference Proceedings* (Vol. 3121, No. 1). AIP Publishing. <https://pubs.aip.org/aip/acp/article-abstract/3121/1/040027/3303066>
- Kaur RP, Jindal MK, Kumar M, Jindal SR, Tuteja S (2022) LineSeg: line segmentation of scanned newspaper documents. *Pattern Anal Appl*, 1–20. <https://doi.org/10.1007/s10044-021-01031-6>

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